

Highlights

Metric Dimension and Local Random Geometric Graphs

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Metric Dimension and Local Random Geometric Graphs

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Abstract

The metric dimension of a graph $G = (V, E)$ is the smallest number of vertices required to uniquely identify all vertices in G based on shortest path distances. In this work, we introduce and study metric dimension on a local random geometric graph (RGG) model in which $n - 1$ vertices are samples from a d -dimensional ℓ^p -ball together with a distinguished central vertex. Each set of vertices induces a partition of the underlying space, connecting metric dimension to the study of a geometric collision probability measuring the likelihood that two independently sampled points are indistinguishable with respect to the chosen vertices.

For $d = 1$, we obtain exact expressions and concentration results for this probability when vertices are randomly distributed and characterize optimal vertex configurations minimizing the probability. For $d > 1$, we derive bounds for $p \in \{1, 2, \infty\}$ and present computational results describing behavior for general p . This analysis suggests corresponding asymptotic behavior for the metric dimension of local RGGs. In particular, we conjecture that

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$\dim(G) = O(\log(n))$ for local RGGs as $n \rightarrow \infty$.

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1. Introduction

Given a graph $G = (V, E)$, $R \subseteq V$ is called *resolving* when, for every pair of distinct vertices $u, v \in V$, there exists some $r \in R$ such that $d(r, u) \neq d(r, v)$ where $d(x, y)$ is the geodesic or shortest path distance between $x, y \in V$. The *metric dimension* of G , denoted $\dim(G)$, is the cardinality of smallest resolving sets on G [1, 2]. Imposing an order on $R = (r_1, \dots, r_{|R|})$, each $v \in V$ has a unique representation $r(v|R) = (d(v, r_1), \dots, d(v, r_{|R|}))$. This makes resolving sets a natural tool for, among other things, applications involving navigation and identification in discrete space [3, 2].

Determining $\dim(G)$ for an arbitrary graph is an NP-hard problem [3]. While exact values, bounds, and asymptotic behavior are known for a wide variety of graph types, including trees [4, 1, 2], grids [3, 5], and wheels [6], relatively little is known about metric dimension on random graphs with the notable exceptions of Erdős-Rényi random graphs [7] and random trees and forests [8]. In this work, we investigate metric dimension in the context of random geometric graphs (RGGs) [9].

A RGG $G(n, r, p, d)$ is defined by n vertices with locations sampled uniformly at random from $[0, 1]^d$ with an edge between two vertices if and only if the p -norm distance between them is less than a fixed radius r . RGGs

serve as models for spatial networks and have been studied extensively in the context of ad hoc wireless networks [10, 11], citation networks [12], and protein-protein interaction networks [13].

We introduce *local random geometric graphs*, a variant of RGGs with a distinguished central vertex and $n - 1$ vertices sampled uniformly from the unit ℓ^p -ball in $\mathbb{R}^d$. This construct serves as a tractable approximation of local neighborhoods in full RGGs and as a probabilistic object of independent interest. In particular, we take advantage of the fact that every subset of vertices induces a partition of the underlying space into regions whose points are indistinguishable with respect to the chosen vertices. We study the probability that two independently sampled points are indistinguishable given such a partition.

In one dimension, we obtain exact expressions for the collision probability under random vertex placement, establish concentration results, and characterize optimal configurations. In higher dimensions, we derive geometric bounds for several values of p and investigate the resulting asymptotic behavior both analytically and computationally. These results suggest corresponding asymptotic behavior for the metric dimension of discrete local random geometric graphs. In addition to local RGGs, we briefly examine full RGGs and compare their behavior with predictions arising from the local theory.

The remainder of the paper is organized as follows. Section 2 introduces local RGGs and develops the continuous partition framework. Section 2.1 presents the analysis of collision probabilities and induced partitions. Section 2.2 investigates implications for metric dimension in discrete local RGGs. Section 3 discusses connections with full random geometric graphs and com-

putational experiments. We conclude with directions for future work and several open problems.

2. Local Random Geometric Graphs

Let $n, d \in \mathbb{N}$ and $\|\cdot\|_p$ denote the ℓ^p -norm on $\mathbb{R}^d$. We write

$$B_p^d = \{x \in \mathbb{R}^d : \|x\|_p \leq 1\}$$

for the unit ℓ^p -ball centered at the origin. A *local random geometric graph*, denoted $G_\ell(n, p, d)$, is constructed by placing a distinguished central vertex c at the origin and sampling $n - 1$ vertices independently and uniformly at random from B_p^d . Two vertices are adjacent when their ℓ^p -distance is at most one.

By construction, c is adjacent to all other vertices and, as a result, the diameter of $G_\ell(n, p, d)$, denoted $\text{diam}(\cdot)$, is at most two. We note that $G_\ell(n, p, d)$ is closely related to ego graphs in full RGGs.

Unless otherwise stated, we restrict our attention to $d \in \{1, 2\}$ and omit the argument d from the notation when the dimension is clear from context.

2.1. Continuous Setting

Let $R = \{r_1, \dots, r_{n-1}\} \subseteq B_p^d$ be a finite set of vertices. For each $x \in B_p^d$, define $\sigma_R(x) = (\lceil \|x - r_1\|_p \rceil, \dots, \lceil \|x - r_{n-1}\|_p \rceil)$. We note that $\|x - r_i\|_p \leq 2$ since $x, r_i \in B_p^d$. As a result, for $x \notin R$, each coordinate of $\sigma_R(x)$ takes either the value 1 or 2. In a local RGG with vertices at points x and r_i , the shortest path distance between these vertices is $\lceil \|x - r_i\|_p \rceil$.

R now induces a natural equivalence relation on B_p^d and an associated partition of the space. In particular, two points $x, y \in B_p^d$ are equivalent

with respect to R , denoted $x \equiv_R y$, when $\sigma_R(x) = \sigma_R(y)$. We call the resulting partition $\mathcal{P}(R)$.

Let X and Y be points drawn uniformly at random from B_p^d . One of the main quantities of interest here is the probability that X and Y are indistinguishable with respect to some set of n vertices R . That is

$$I(R) = P(X \equiv_R Y) = \sum_{A \in \mathcal{P}(R)} \mu(A)^2$$

where μ is the normalized Lebesgue measure on B_p^d .

Throughout the paper we distinguish between deterministic and random vertex sets. When R_n is sampled independently and uniformly from B_p^d , we write $I_n = I(R_n)$. In the continuous setting, our goal is to understand the behavior of I_n , including its expectation, variance, concentration, and asymptotic scaling, for different dimensions and values of p .

2.1.1. One Dimension

When $d = 1$, $B_p^1 = [-1, 1]$ and the resulting partition is independent of the choice of $p \geq 1$. Each vertex $r_i \in [-1, 1]$ partitions the interval into two subintervals corresponding to points whose distance from r_i is at most one and greater than one, respectively (see Figure 1, top).

Let $R_n = \{r_1, \dots, r_{n-1}\}$ where the r_i are sampled independently and uniformly from $[-1, 1]$. Ordering the vertices so that $-1 < r_{(1)} < \dots < r_{(n-1)} < 1$, the partition $\mathcal{P}(R_n)$ consists of n intervals. Define the interval lengths $X_i = r_{(i)} - r_{(i-1)}$ for $1 \leq i \leq n$ where $r_{(0)} = -1$ and $r_{(n)} = 1$. The distribution of these lengths can be analyzed in several ways [14, 15].

Here and in subsequent sections, we will view these lengths as being drawn

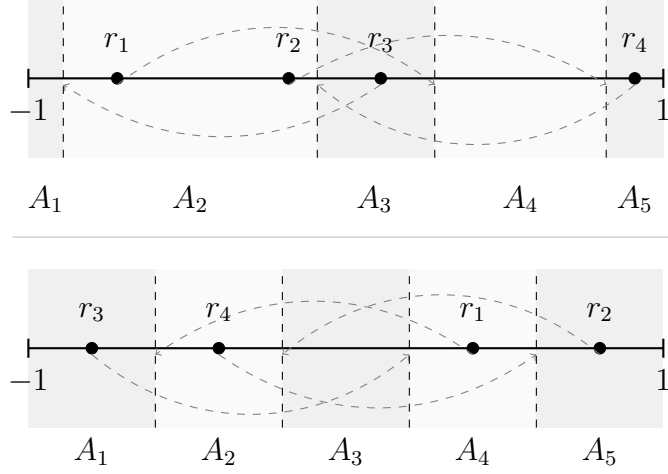


Figure 1: One-dimensional partitions induced by four vertices. Regions are labeled and shaded. Curved dashed lines connect r_i with their associated boundary. Top) a random set of vertices. Bottom) an optimal set of vertices R^* .

from a Dirichlet distribution [16]. In particular,

$$(X_1, \dots, X_n) \sim 2 \cdot \text{Dirichlet}(1, \dots, 1)$$

and, as a result,

$$\mathbb{E}[X_i] = \frac{2}{n} \quad \mathbb{E}[X_i^2] = \frac{8}{n(n+1)}$$

for every $1 \leq i \leq n$.

Since two independently sampled points are indistinguishable if and only if they lie in the same interval of $\mathcal{P}(R_n)$, we obtain

$$I_n = \sum_{i=1}^n \left(\frac{X_i}{2} \right)^2$$

where the factor of $\frac{1}{2}$ accounts for the normalized Lebesgue measure on $[-1, 1]$.

The expectation and variance of I_n admit simple closed forms. In particular,

$$\begin{aligned}\mathbb{E}[I_n] &= \sum_{i=1}^n \mathbb{E}\left[\left(\frac{X_i}{2}\right)^2\right] \\ &= \frac{n}{4} \mathbb{E}[X_1^2] \\ &= \frac{n}{4} \cdot \frac{8}{n(n+1)} \\ &= \frac{2}{n+1}.\end{aligned}$$

For the variance, we compute the second moment of I_n . Using standard moments of the Dirichlet distribution,

$$\mathbb{E}\left[\left(\frac{X_i}{2}\right)^4\right] = \frac{24}{n(n+1)(n+2)(n+3)}$$

and, for $i \neq j$,

$$\mathbb{E}\left[\left(\frac{X_i X_j}{4}\right)^2\right] = \frac{4}{n(n+1)(n+2)(n+3)}.$$

Then we have

$$\begin{aligned}\mathbb{E}[I_n^2] &= n \mathbb{E}\left[\left(\frac{X_1}{2}\right)^4\right] + n(n-1) \mathbb{E}\left[\left(\frac{X_1 X_2}{4}\right)^2\right] \\ &= \frac{24 + 4(n-1)}{(n+1)(n+2)(n+3)} \\ &= \frac{4(n+5)}{(n+1)(n+2)(n+3)}.\end{aligned}$$

Since $\mathbb{E}[I_n] = \frac{2}{n+1}$,

$$\begin{aligned}\text{Var}[I_n] &= \mathbb{E}[I_n^2] - \mathbb{E}[I_n]^2 \\ &= \frac{4(n+5)}{(n+1)(n+2)(n+3)} - \frac{4}{(n+1)^2} \\ &= \frac{4(n-1)}{(n+1)^2(n+2)(n+3)}\end{aligned}$$

and we have that $\text{Var}(I_n) = \Theta(n^{-3})$.

Applying Chebyshev's inequality immediately yields a concentration result. For every $\delta > 0$,

$$\Pr\left(\left|I_n - \frac{2}{n+1}\right| \geq \delta\right) \leq \frac{4(n-1)}{\delta^2(n+1)^2(n+2)(n+3)}.$$

In particular, $I_n - \frac{2}{n+1} \rightarrow 0$ in probability.

By the Cauchy-Schwarz inequality, $I(R) = \sum_{A \in \mathcal{P}(R)} \mu(A)^2$ is minimized when the induced partition consists of n intervals of equal length and every partition has normalized measure $\frac{1}{n}$. In particular, $\sum_{A \in \mathcal{P}(R)} \mu(A) = 1$, so

$$1 = \left(\sum_{A \in \mathcal{P}(R)} \mu(A)\right)^2 \leq n \sum_{A \in \mathcal{P}(R)} \mu(A)^2 = nI(R)$$

and $I(R) \geq \frac{1}{n}$.

One optimal configuration is $R^* = \{r_1, \dots, r_{n-1}\}$ such that

$$r_i = \begin{cases} \frac{2i}{n}, & i < \frac{n}{2} \\ \frac{2i}{n} - 2, & i \geq \frac{n}{2} \end{cases}$$

and $I(R^*) = \frac{1}{n}$ (see Figure 1, bottom). Thus, random vertex placement is asymptotically within a factor of two of the optimal configuration.

Finally, to develop additional intuition in this setting, we consider the probability of a collision between two randomly sampled points given the distance between them.

Lemma 2.1. *Let X and Y be independent uniformly distributed points in $[-1, 1]$. Then*

$$\Pr(X \equiv_R Y \mid |X - Y| = t) = \left(1 - \frac{t}{2}\right)^{n-1}.$$

Proof. Suppose $|X - Y| = t$. The partition induced by r_i has a unique boundary in $[-1, 1]$ at $r_i - 1$ if $r_i > 0$ and at $r_i + 1$ if $r_i < 0$. Since r_i is uniform on the interval, so is this boundary. X and Y are distinguished by r_i precisely when this boundary lies between them, which occurs with probability $\frac{t}{2}$. Hence, a single point fails to distinguish the points with probability $1 - \frac{t}{2}$. With $n - 1$ independent points r_i and their associated boundaries, the probability that X and Y fall in the same region is $(1 - \frac{t}{2})^{n-1}$. $\square$

2.1.2. Two Dimensions

The situation becomes somewhat more complex when $d = 2$. Now, the shapes of the regions defined by $R \subseteq B_p^2$ and the associated ℓ^p -balls are intricate and depend heavily on p . In this section, we focus our attention on $p = \infty$ where ℓ^p -balls are axis-aligned squares (see Figure 2), though note that our results apply equally when $p = 1$ after appropriate rotation and scaling.

We begin by constructing a refinement of the partition induced by R , leading to a lower bound on $I(R)$.

Definition 2.2 (Lower Bound Refinement). *Let $R = \{r_1, \dots, r_{n-1}\} \subseteq B_\infty^2$ such that $r_i = (a_i, b_i)$. For each r_i , the ℓ^∞ -ball of radius one centered at r_i is the axis-aligned square $[a_i - 1, a_i + 1] \times [b_i - 1, b_i + 1]$. Extend the horizontal and vertical sides of these squares across B_∞^2 . Let $\tilde{\mathcal{P}}(R)$ denote the rectangular partition of B_∞^2 induced by these extended line segments. Define*

$$L(R) = \sum_{A \in \tilde{\mathcal{P}}(R)} \mu(A)^2,$$

where μ denotes normalized Lebesgue measure on B_∞^2 . See Figure 3 for an illustrative example.

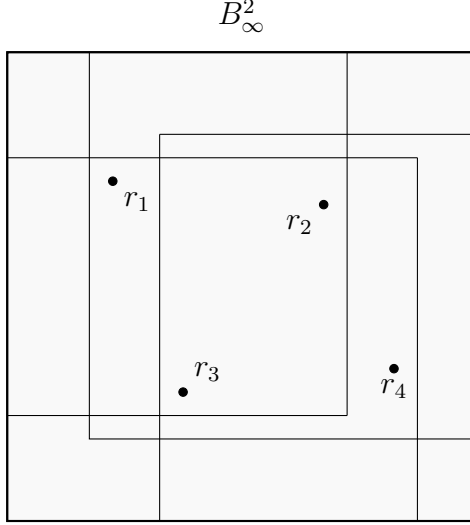


Figure 2: Clipped ℓ^∞ -balls of radius one centered at vertices in B_∞^2 .

Lemma 2.3. *For every finite $R \subseteq B_\infty^2$, $L(R) \leq I(R)$.*

Proof. The partition $\tilde{\mathcal{P}}(R)$ is a refinement of $\mathcal{P}(R)$ induced by the distance vectors to the elements of R . Indeed, if two points lie in the same region of $\tilde{\mathcal{P}}(R)$, they must also lie in the same region of $\mathcal{P}(R)$. As a result, every region of $\mathcal{P}(R)$ is a union of regions of $\tilde{\mathcal{P}}(R)$ and, since splitting a set into smaller pieces can only decrease the sum of squares, we have

$$\sum_{A \in \tilde{\mathcal{P}}(R)} \mu(A)^2 \leq \sum_{A \in \mathcal{P}(R)} \mu(A)^2$$

and $L(R) \leq I(R)$. □

After this refinement, the two-dimensional problem separates into independent one-dimensional spacing problems in the vertical and horizontal directions. The lower bound is essentially the product of two copies of the one-dimensional collision probability.

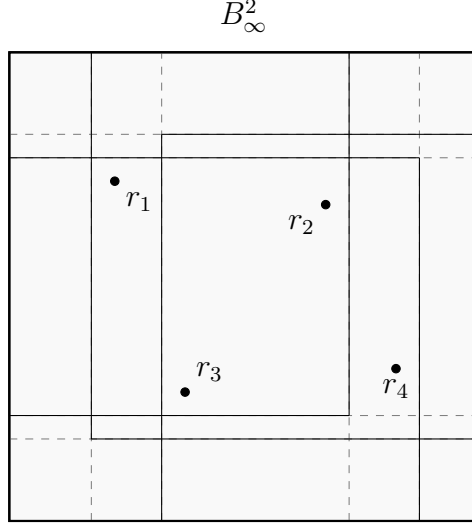


Figure 3: The lower-bound refinement obtained by extending the horizontal and vertical sides of the clipped ℓ^∞ -balls across B_∞^2 . Extensions are dashed.

Lemma 2.4 (Lower bound). *Let R_n consist of $n - 1$ independent uniformly distributed vertices in B_∞^2 and define $L_n = L(R_n)$. Then*

$$\mathbb{E}[L_n] = \frac{4}{(n+1)^2}.$$

Moreover,

$$\text{Var}(L_n) = \Theta(n^{-5})$$

and

$$L_n - \frac{4}{(n+1)^2} \rightarrow 0$$

in probability. In particular,

$$\mathbb{E}[I_n] \geq \frac{4}{(n+1)^2}.$$

Proof. Each vertex in R contributes one vertical and one horizontal cut to the refined rectangular partition. Thus the horizontal and vertical spacings

satisfy

$$(X_1, \dots, X_n) \sim 2 \text{ Dirichlet}(1, \dots, 1), \quad (Y_1, \dots, Y_n) \sim 2 \text{ Dirichlet}(1, \dots, 1).$$

Then we have

$$L_n = \sum_{i=1}^n \sum_{j=1}^n \left(\frac{X_i Y_j}{4} \right)^2 = \left(\sum_{i=1}^n \left(\frac{X_i}{2} \right)^2 \right) \left(\sum_{j=1}^n \left(\frac{Y_j}{2} \right)^2 \right).$$

Using results from the one-dimensional case, we have

$$\begin{aligned} \mathbb{E}[L_n] &= \mathbb{E} \left[\left(\sum_{i=1}^n \left(\frac{X_i}{2} \right)^2 \right) \left(\sum_{j=1}^n \left(\frac{Y_j}{2} \right)^2 \right) \right] \\ &= \mathbb{E} \left[\sum_{i=1}^n \left(\frac{X_i}{2} \right)^2 \right]^2 \\ &= \left(\frac{2}{n+1} \right)^2 \\ &= \frac{4}{(n+1)^2}. \end{aligned}$$

The variance of L_n follows from the one-dimensional computation. Indeed, let $A_n = \sum_{i=1}^n \left(\frac{X_i}{2} \right)^2$ and $B_n = \sum_{j=1}^n \left(\frac{Y_j}{2} \right)^2$. Then $L_n = A_n B_n$ where A_n and B_n are independent. Since $\mathbb{E}[A_n] = \Theta(\frac{1}{n})$ and $\text{Var}(A_n) = \Theta(\frac{1}{n^3})$, we find that

$$\begin{aligned} \text{Var}[L_n] &= \text{Var}[A_n B_n] \\ &= \mathbb{E}[A_n]^2 \text{Var}[B_n] + \mathbb{E}[B_n]^2 \text{Var}[A_n] + \text{Var}[A_n] \text{Var}[B_n] \\ &= 2\mathbb{E}[A_n]^2 \text{Var}[A_n] + \text{Var}[A_n]^2 \\ &= \Theta\left(\frac{1}{n^5}\right) \end{aligned}$$

and, applying Chebyshev's inequality,

$$L_n - \frac{4}{(n+1)^2} \rightarrow 0$$

in probability. Finally, $L_n \leq I_n$ pointwise, so

$$\mathbb{E}[I_n] \geq \mathbb{E}[L_n] = \frac{4}{(n+1)^2}.$$

□

Next, we construct a coarsening of the partition giving an upper bound on $I(R)$.

Definition 2.5 (Upper bound coarsening). *Let*

$$Q_1 = [0, 1] \times [0, 1], \quad Q_2 = [-1, 0] \times [0, 1],$$

$$Q_3 = [-1, 0] \times [-1, 0], \quad Q_4 = [0, 1] \times [-1, 0]$$

denote the four quadrants of B_∞^2 . Within each quadrant, only keep boundary segments of $\mathcal{P}(R)$ which connect two opposite sides of the quadrant. For example, in Q_1 , vertices in Q_2 induce full vertical cuts at $x = a + 1$ and vertices in Q_4 induce full horizontal cuts at $y = b + 1$. All other boundary segments in Q_1 are ignored.

Let $\mathcal{Q}_i(R)$ denote the resulting rectangular partition of Q_i , and define

$$U_i(R) = \sum_{A \in \mathcal{Q}_i(R)} \mu(A)^2.$$

Finally,

$$U(R) = \sum_{i=1}^4 U_i(R).$$

See Figure 4 for an illustrative example.

For the upper bound, we discard many boundary segments resulting in a coarsening of the partition and larger collision probabilities. Since this is performed for each quadrant separately, Cauchy-Schwarz is used to bound the contribution of regions overlapping multiple quadrants.

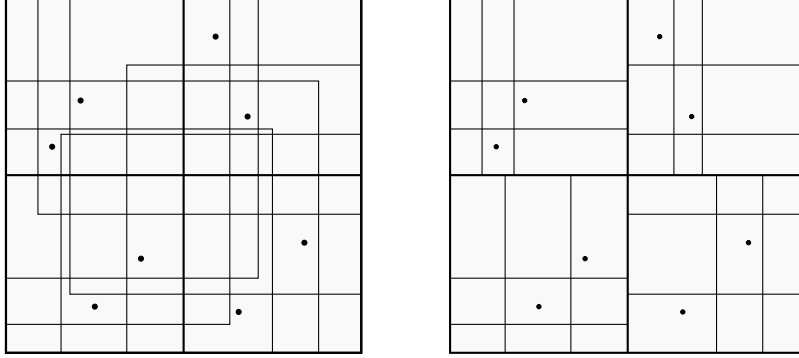


Figure 4: Upper-bound coarsening in B_∞^2 . The left panel shows the clipped ℓ^∞ -balls, while the right panel keeps only the full horizontal and vertical cuts within each quadrant.

Lemma 2.6. *For every finite $R \subseteq B_\infty^2$, $I(R) \leq 4U(R)$.*

Proof. Let $\mathcal{P}(R) = \{S_\alpha\}$ denote the partition of B_∞^2 induced by the distance vectors to the vertices in R . For each quadrant Q_i , define

$$I_i(R) = \sum_{\alpha} \mu(S_\alpha \cap Q_i)^2.$$

Since

$$\mu(S_\alpha) = \sum_{i=1}^4 \mu(S_\alpha \cap Q_i),$$

the Cauchy-Schwarz inequality gives

$$\mu(S_\alpha)^2 = \left(\sum_{i=1}^4 \mu(S_\alpha \cap Q_i) \right)^2 \leq 4 \sum_{i=1}^4 \mu(S_\alpha \cap Q_i)^2.$$

Summing over all regions gives

$$I(R) = \sum_{\alpha} \mu(S_\alpha)^2 \leq 4 \sum_{i=1}^4 I_i(R).$$

Consider a fixed quadrant Q_i . The partition of Q_i induced by $S_\alpha \cap Q_i$ refines the coarsening used to define $U_i(R)$. In particular, merging regions

inside Q_i can only increase the sum of squares, and we have

$$I_i(R) \leq U_i(R)$$

where $U_i(R)$ is the contribution of Q_i to $U(R)$. So

$$I(R) \leq 4 \sum_{i=1}^4 U_i(R) = 4U(R).$$

□

Lemma 2.7. *For a fixed quadrant Q_i with V vertical and H horizontal cuts after coarsening,*

$$\mathbb{E}[U_i \mid V, H] = \frac{1}{4(V+2)(H+2)}.$$

Proof. With V vertical and H horizontal cuts, the spacings satisfy

$$(X_1, \dots, X_{V+1}) \sim \text{Dirichlet}(1, \dots, 1), \quad (Y_1, \dots, Y_{H+1}) \sim \text{Dirichlet}(1, \dots, 1)$$

and, since these are independent of one another, we have

$$\mathbb{E} \left[\sum_{j=1}^{V+1} X_j^2 \right] = \frac{2}{V+2} \quad \mathbb{E} \left[\sum_{j=1}^{H+1} Y_j^2 \right] = \frac{2}{H+2}.$$

Then

$$\begin{aligned} \mathbb{E}[U_i \mid V, H] &= \mathbb{E} \left[\left(\sum_{j=1}^{V+1} \left(\frac{X_j}{2} \right)^2 \right) \left(\sum_{j=1}^{H+1} \left(\frac{Y_j}{2} \right)^2 \right) \right] \\ &= \frac{1}{4(V+2)(H+2)}. \end{aligned}$$

□

The coarsened partition in each quadrant is controlled by the number of full vertical and horizontal cuts. Since these counts are binomial, they will tend to concentrate around the mean and each quadrant looks similar to the situation in Lemma 2.4.

Lemma 2.8 (Upper bound). *Let R_n consist of $n - 1$ independent uniformly distributed vertices in B_∞^2 , and define $U_n = U(R_n)$. Then*

$$\mathbb{E}[U_n] = O\left(\frac{1}{n^2}\right).$$

As a result,

$$\mathbb{E}[I_n] = O\left(\frac{1}{n^2}\right)$$

and $I_n \rightarrow 0$ in probability.

Proof. For a fixed quadrant Q_i , let V and H be the numbers of vertical and horizontal cuts in Q_i after coarsening. Note that V and H come from a Multinomial($n - 1, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$) distribution. By Lemma 2.7,

$$\mathbb{E}[U_i] = \frac{1}{4} \mathbb{E} \left[\frac{1}{(V + 2)(H + 2)} \right].$$

To evaluate this expectation, we use the integral identity

$$\frac{1}{k + 2} = \int_0^1 x^{k+1} dx.$$

In particular,

$$\frac{1}{(V + 2)(H + 2)} = \int_0^1 \int_0^1 x^{V+1} y^{H+1} dx dy$$

and, applying Fubini's theorem,

$$\mathbb{E} \left[\frac{1}{(V + 2)(H + 2)} \right] = \int_0^1 \int_0^1 xy \mathbb{E}[x^V y^H] dx dy.$$

A single randomly sampled vertex contributes a vertical cut to Q_i with probability $\frac{1}{4}$, a horizontal cut with probability $\frac{1}{4}$, and no cut with probability $\frac{1}{2}$. Then

$$\mathbb{E}[x^V y^H] = \left(\frac{2 + x + y}{4} \right)^{n-1}$$

and

$$\mathbb{E}[U_i] = \frac{1}{4} \int_0^1 \int_0^1 xy \left(\frac{2+x+y}{4} \right)^{n-1} dx dy.$$

Since $x, y \leq 1$,

$$\frac{2+x+y}{4} = 1 - \frac{(1-x) + (1-y)}{4}$$

and, using $1-t \leq e^{-t}$,

$$\left(\frac{2+x+y}{4} \right)^{n-1} \leq e^{-\frac{n-1}{4}(1-x)} e^{-\frac{n-1}{4}(1-y)}.$$

As a result,

$$\begin{aligned} \mathbb{E}[U_i] &\leq \frac{1}{4} \left(\int_0^1 e^{-\frac{n-1}{4}(1-x)} dx \right)^2 \\ &= \frac{1}{4} \left(\frac{4}{n-1} (1 - e^{-\frac{n-1}{4}}) \right)^2 \\ &= \frac{4}{(n-1)^2} (1 - e^{-\frac{n-1}{4}})^2 \\ &= O\left(\frac{1}{n^2}\right) \end{aligned}$$

Summing over the four quadrants,

$$\mathbb{E}[U_n] = \sum_{i=1}^4 \mathbb{E}[U_i] = O\left(\frac{1}{n^2}\right).$$

Since $I_n \leq 4U_n$ pointwise,

$$\mathbb{E}[I_n] \leq 4\mathbb{E}[U_n] = O\left(\frac{1}{n^2}\right).$$

Finally, for every $\varepsilon > 0$, Markov's inequality gives

$$\mathbb{P}(I_n > \varepsilon) \leq \frac{\mathbb{E}[I_n]}{\varepsilon}.$$

Since $\mathbb{E}[I_n] = O\left(\frac{1}{n^2}\right) \rightarrow 0$, $\mathbb{P}(I_n > \varepsilon) \rightarrow 0$ and $I_n \rightarrow 0$ in probability. $\square$

Finally, we combine these bounds to show that I_n has order $1/n^2$ in B_∞^2 . The result follows directly from Lemmas 2.4 and 2.8.

Theorem 2.9. *For $d = 2$ and $p = \infty$,*

$$\mathbb{E}[I_n] = \Theta\left(\frac{1}{n^2}\right).$$

Moreover, $I_n \rightarrow 0$ in probability.

2.1.3. Experimental Results

2.2. Discrete Setting

3. Random Geometric Graphs

4. Future Work

5. Conclusion

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